

SE3082

AWS EC2 Lab - Screenshots and Calculations

Lab 06

Exercise 3: Pragma omp (openmp) master code

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On instance with 2 vCPUs (t3.micro)

Nano - creating the pragma omp master code

```
GNU nano 6.2          ex3_master.c
#include <omp.h>
#include <stdio.h>

int main() {

#pragma omp parallel
{
    int th = omp_get_thread_num();
    printf("thread # %d\n", th);

#pragma omp master
{
    printf("Inside Master # %d\n", omp_get_thread_num());
    printf("Exiting Master \n");
}
    printf("Hi again from thread # %d\n", omp_get_thread_num());
}

    return 0;
}
```

Outputs from running the pragma omp master code.

```
[ec2-user@ip-172-31-17-201 ~]$ gcc -fopenmp ex3_master.c -o ex3_master.o
[ec2-user@ip-172-31-17-201 ~]$ ./ex3_master.o
thread # 1
Hi again from thread # 1
thread # 0
Inside Master # 0
Exiting Master
Hi again from thread # 0
[ec2-user@ip-172-31-17-201 ~]$ ./ex3_master.o
thread # 1
Hi again from thread # 1
thread # 0
Inside Master # 0
Exiting Master
Hi again from thread # 0
[ec2-user@ip-172-31-17-201 ~]$ ./ex3_master.o
thread # 1
Hi again from thread # 1
thread # 0
Inside Master # 0
Exiting Master
Hi again from thread # 0
[ec2-user@ip-172-31-17-201 ~]$ ./ex3_master.o
thread # 0
Inside Master # 0
Exiting Master
Hi again from thread # 0
thread # 1
Hi again from thread # 1
```

How the `pragma omp master` code works in the program.

`#pragma omp master` restricts execution of the block to the **master thread** specifically, **thread 0**, and only thread 0. Unlike `single`, which lets whichever thread arrives first claim the block (any thread number possible), `master` always assigns the work to thread 0 by definition, regardless of arrival order or scheduling.

Another key difference: `master` has no implicit barrier, so other threads don't wait for thread 0 to finish the `master` block before continuing to "Hi again", meaning we might occasionally see some "Hi again" lines interleave with or even appear before "Inside Master" or "Exiting Master" depending on scheduling timing, unlike Ex2's `single` which enforced a clean separation via its implicit barrier.